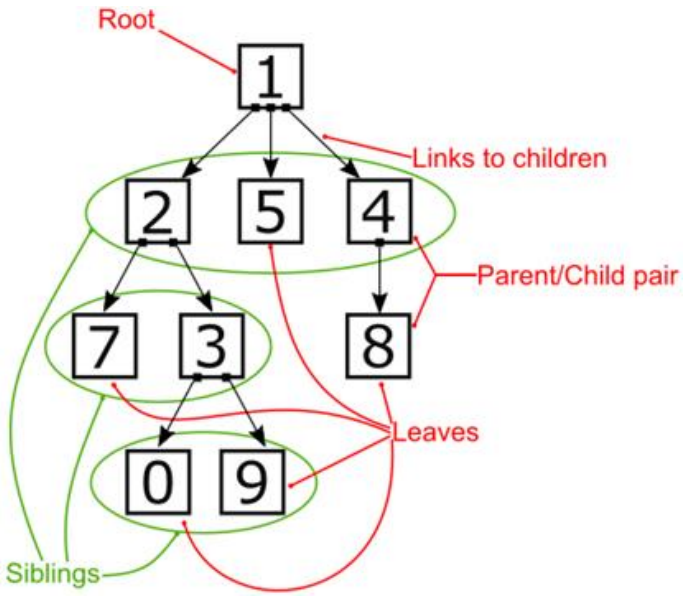


Figure C.9 An example of a *tree*. The top-to-bottom arrows represent the links between a node and its children which are always directed from parent to children. Every node has one and only one parent. Nodes that share the same parent are called *siblings*. There is no explicit link between siblings, but one node's siblings can only be reached through their common parent. Nodes with no children are called *leaves*.

Before moving on, let's review the terminology used when dealing with trees.

- If a node `x` has two children `y` and `z`, `x` is the parent of these nodes. So, the following is true: `parent(y) == parent(z) == x`. In figure C.9, The node labeled `1` is the parent of the nodes labeled `4`, `5` and `2`.
- Being `x`, `y` and `z` defined as above, `y` and `z` are called *siblings*.
- The *root* of a tree is the only node in a tree that has no parent.
- An ancestor of a node `x` is any node in the path from the tree's root to `x`. In other words, either `parent(x)`, `parent(parent(x))` and so on... In figure C.9, `2` is an ancestor of `7`, `3`, `0` and `9`.
- A *leaf* is any node that has no children. In other words, the node has two empty subtrees as children.
- A fundamental characteristic of a tree is its *height*, defined as the length of the longest path from the root to a leaf. The height of the